

Executive Scenario Report

Predicted Load	Uncertainty	Financial Exposure	Grid Status
120.3 MWh	+/- 7.0 MWh	\$16,123	Elevated

Strategic Overview

This is a short-term operational energy forecast predicting a value of approximately 120.32 units for the specified hour, with an uncertainty range of about +/- 7 units. The ensemble model heavily weights the CatBoost prediction, while recent historical lags (12h and 24h) are negatively influencing the current forecast, suggesting a potential downward adjustment from previous periods. This forecast is critical for immediate operational planning and resource allocation.

Critical Risk Factors

- The significant uncertainty margin (approximately +/- 7 units) could lead to operational deviations and resource misallocation if not properly accounted for in real-time planning.
- Heavy reliance on the CatBoost model (65% weight) introduces vulnerability; if this specific model experiences an unforeseen error or bias, the overall ensemble prediction could be compromised.
- Negative contributions from recent historical lags (lag_12h, lag_24h) suggest a potential downward adjustment that might be overly sensitive to short-term fluctuations, risking underestimation if conditions stabilize or rebound unexpectedly.

Actionable Recommendations

- Implement real-time monitoring of actual energy consumption against this forecast, establishing clear thresholds for deviation to trigger immediate operational adjustments.
- Conduct a deeper analysis into the negative contributions of lag features (12h and 24h) to understand underlying drivers and validate if the model's downward adjustment is justified or if external factors are at play.
- Regularly review and optimize the ensemble model weights, particularly for the heavily weighted CatBoost component, to ensure its continued accuracy and robustness across varying operational conditions.

Input Feature Context

Hour:	12
Day Of Week:	2
Month:	1

Is Weekend:	0
Lag 24H:	120
Lag 12H:	118
Roll Mean 24H:	125
Roll Std 24H:	4